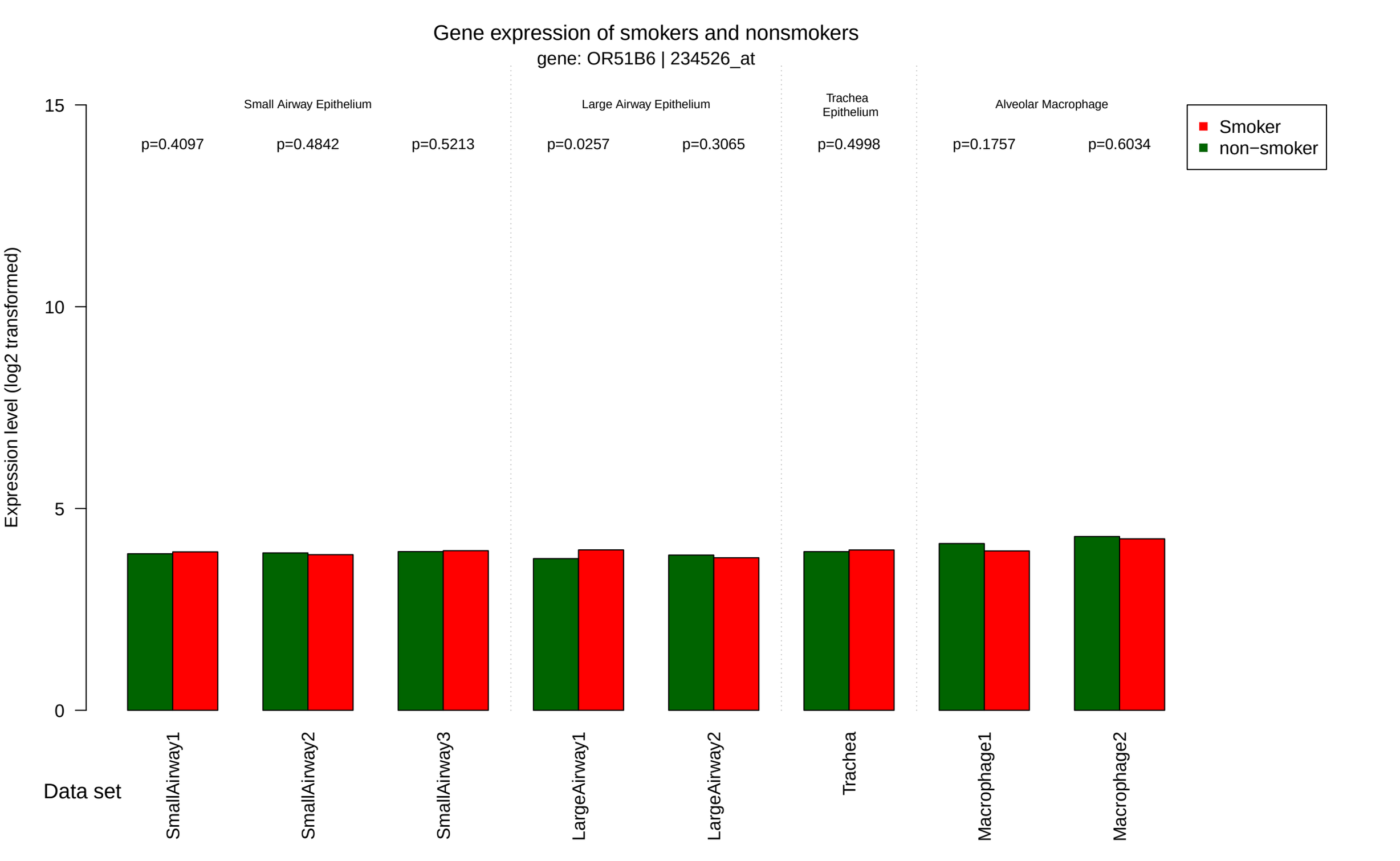
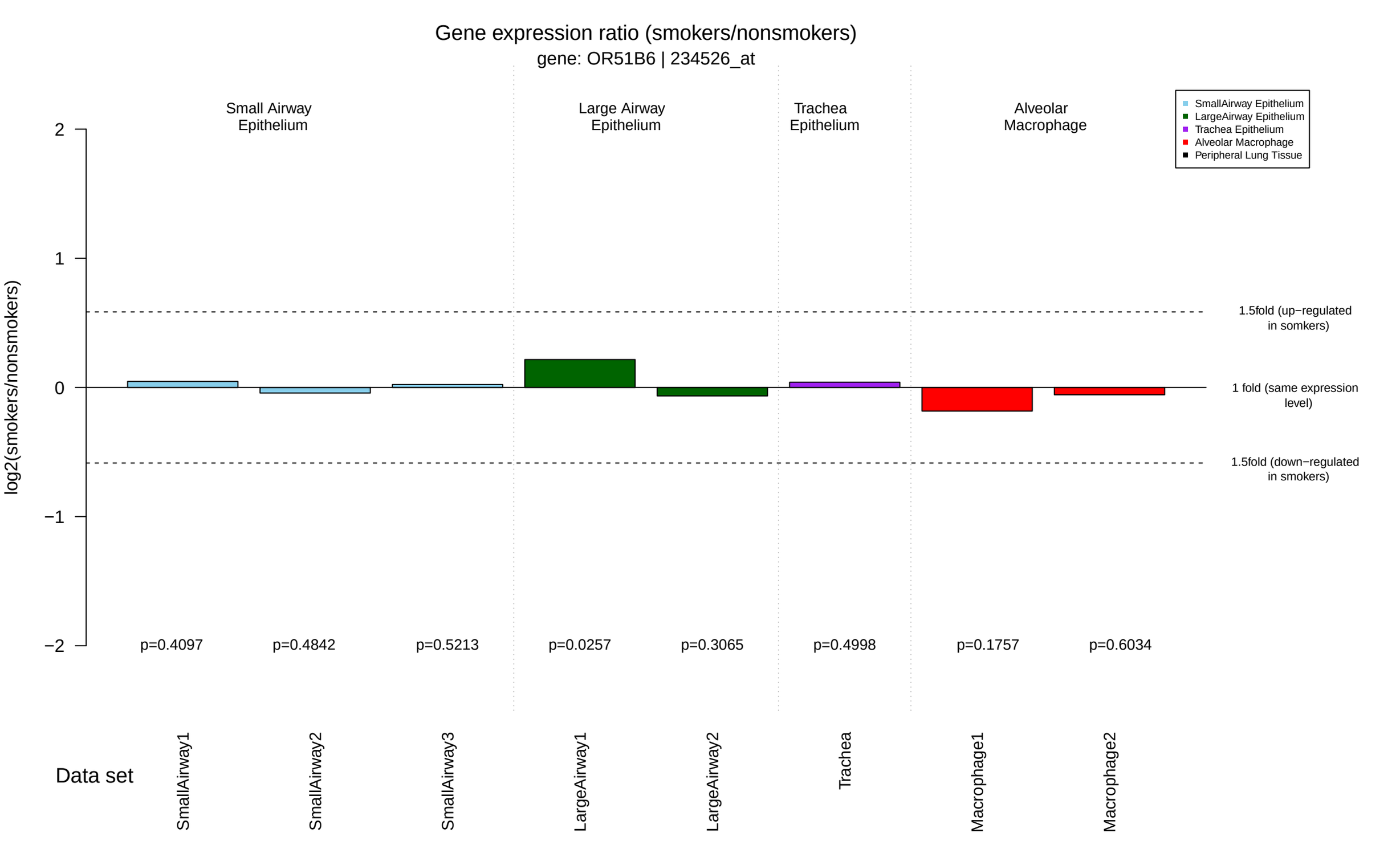
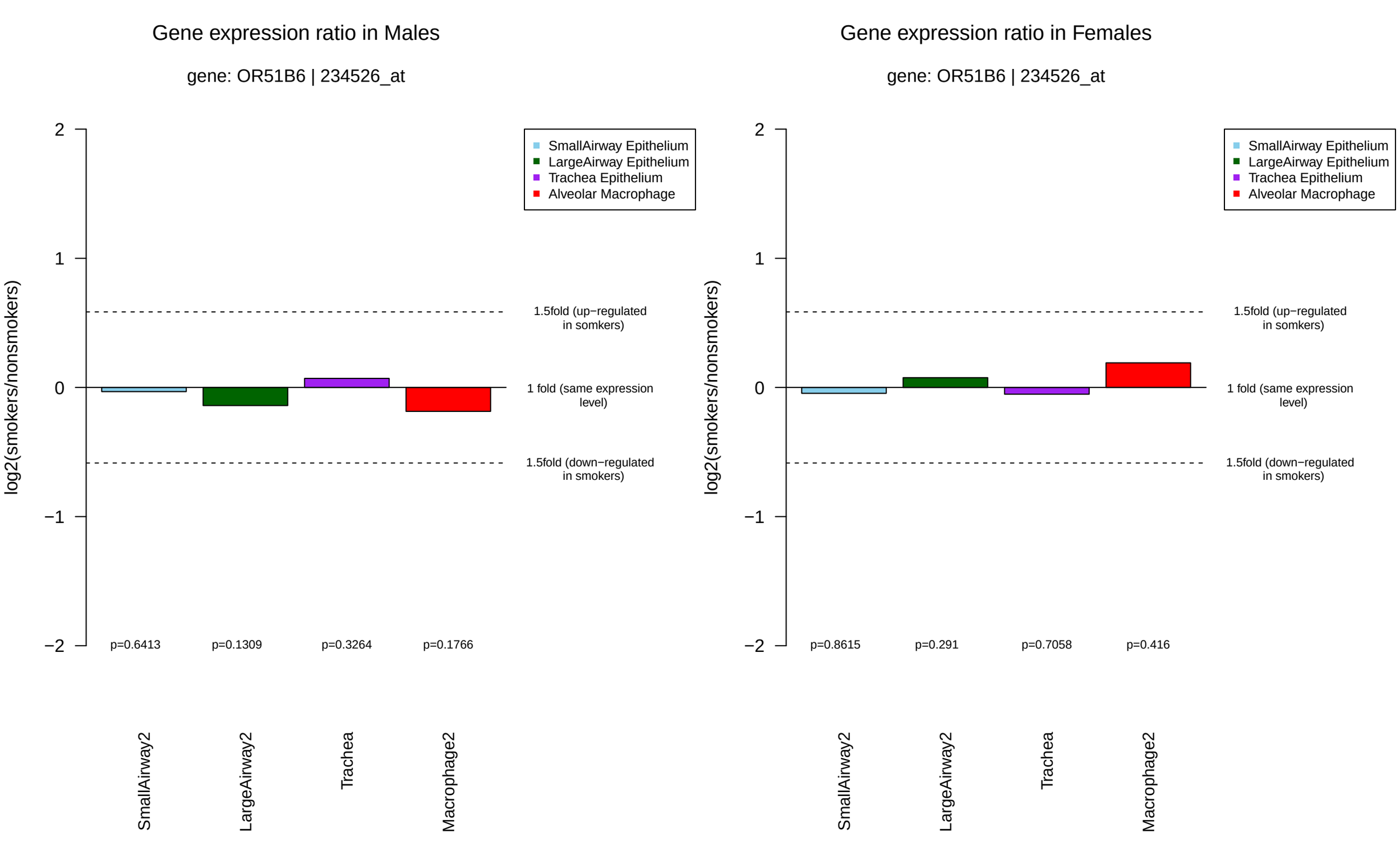
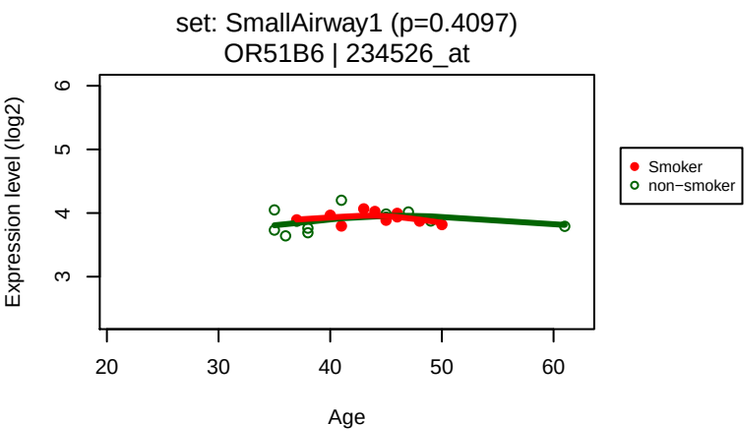
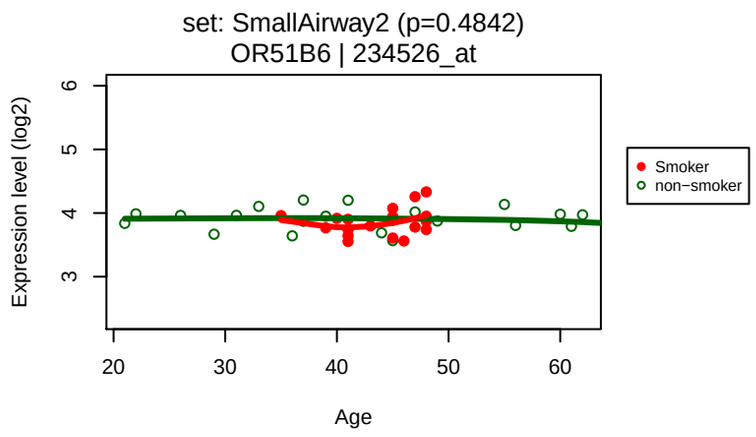
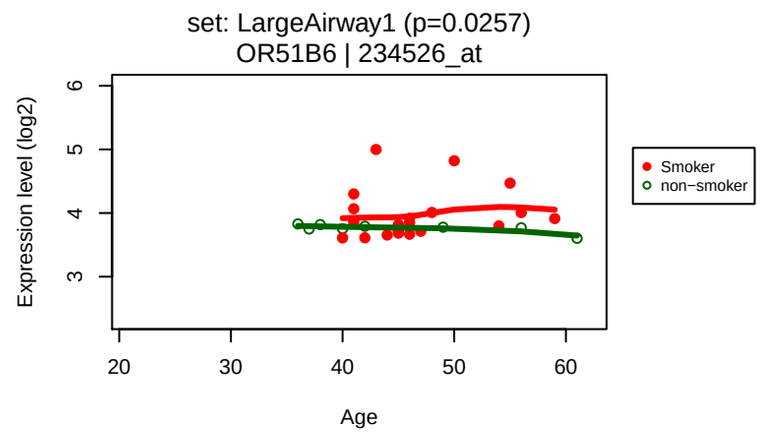

Gene expression of smokers and nonsmokers
gene: OR51B6 | 234526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4097
p=0.4842
p=0.5213
p=0.0257
p=0.3065
p=0.4998
p=0.1757
p=0.6034
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

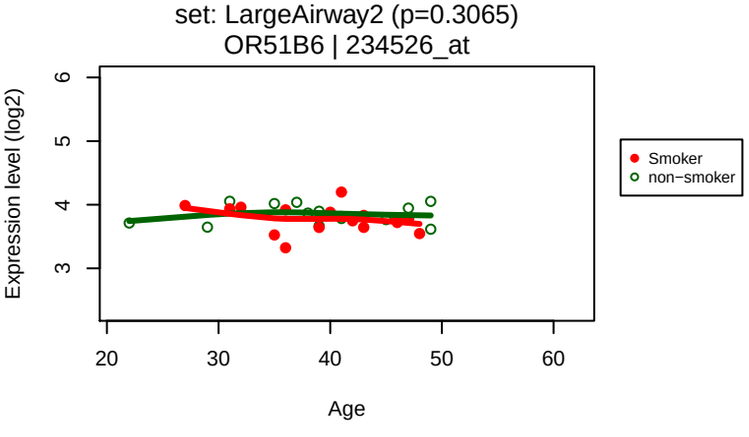

Gene expression ratio (smokers/nonsmokers)
gene: OR51B6 | 234526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4097
p=0.4842
p=0.5213
p=0.0257
p=0.3065
p=0.4998
p=0.1757
p=0.6034
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

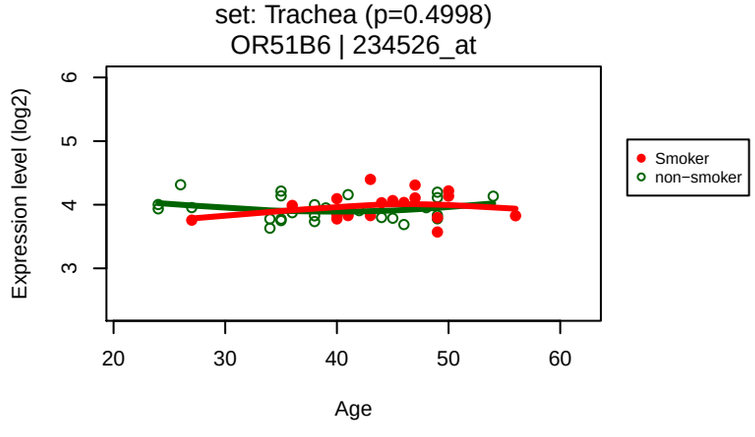

Gene expression ratio in Males
gene: OR51B6 | 234526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6413
p=0.1309
p=0.3264
p=0.1766
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51B6 | 234526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8615
p=0.291
p=0.7058
p=0.416
SmallAirway2
LargeAirway2
Trachea
Macrophage2

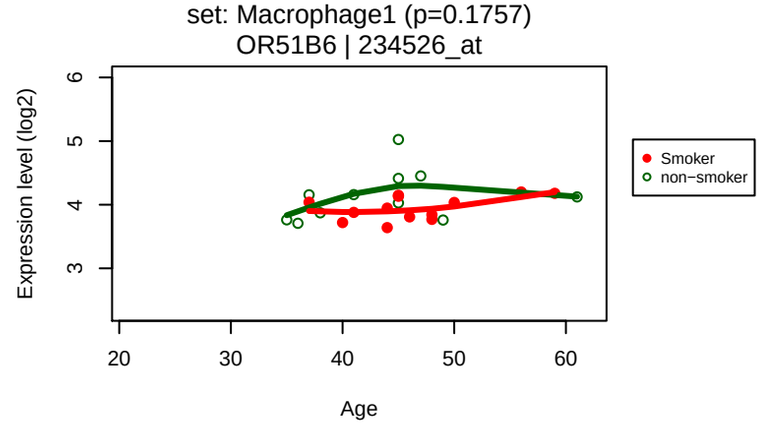

set: SmallAirway1 (p=0.4097)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker

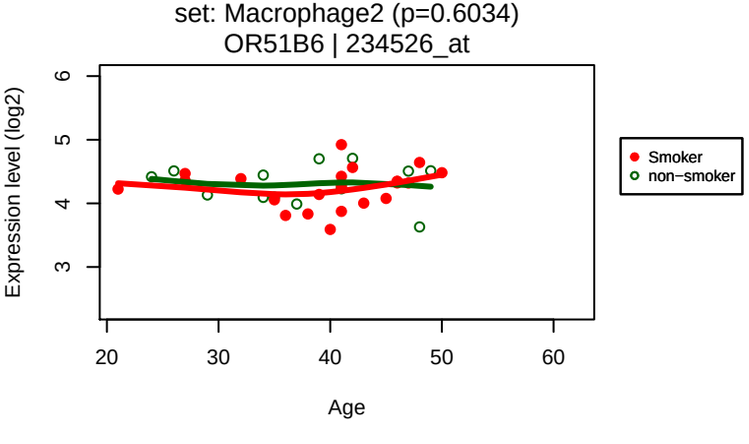

set: SmallAirway2 (p=0.4842)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker

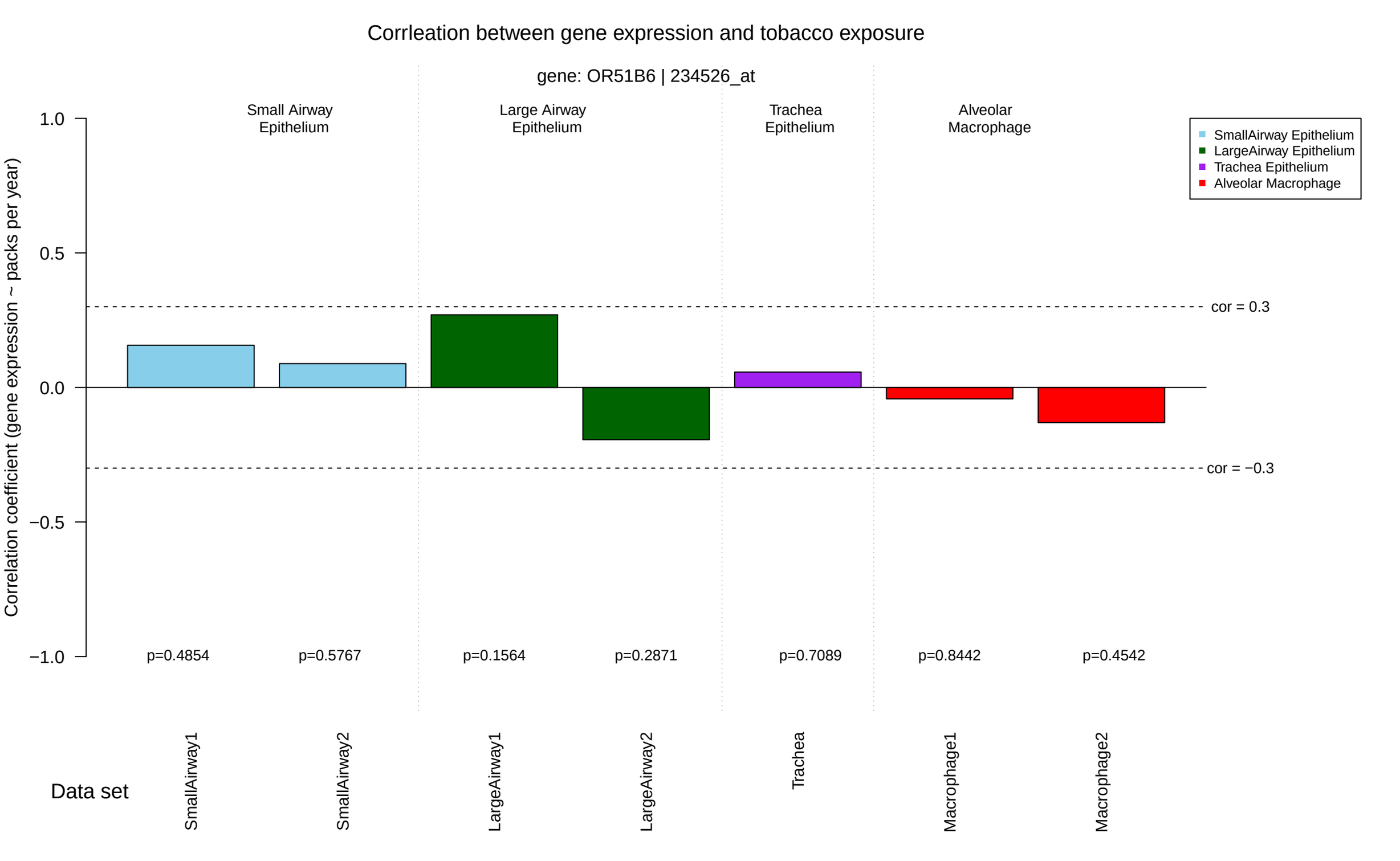

set: LargeAirway1 (p=0.0257)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3065)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4998)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1757)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6034)
OR51B6 | 234526_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51B6 | 234526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4854
p=0.5767
p=0.1564
p=0.2871
p=0.7089
p=0.8442
p=0.4542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2